
Addressing Issues in Automated Skin Lesion Classification Using Convolutional Neural Networks

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Abstract

Skin cancer is one of the most prevalent cancers in the world. Automated classification of skin lesions from dermoscopic images has the potential to improve early diagnosis, which is crucial for effective treatment. In this study, we trained a convolutional neural network for skin lesion classification using a pretrained ResNet-50 architecture with full fine-tuning. We addressed two major challenges commonly encountered in this task: class imbalance and overfitting. Three methods for handling class imbalance were evaluated - balanced batch sampling, weighted cross entropy loss, and focal loss - but none demonstrated superior performance. To mitigate overfitting, strong regularization techniques, including data augmentation, dropout, and weight decay, were applied. The final model achieved a balanced accuracy of 74% on the test set.

Keywords Skin lesion classification · CNN · Transfer learning · Class imbalance · Overfitting

1 Introduction

Skin cancer is one of the most common cancers worldwide [1], and early detection is crucial for effective treatment. Automated classification of skin lesions from dermoscopic images has the potential to assist clinicians in screening and diagnosis. However, this task remains challenging due to the limited availability of labeled data and the high class imbalance typically found in existing datasets, where benign lesions are far more prevalent than malignant ones. Moreover, variations in image quality, lighting, and lesion appearance further complicate model training and generalization [2].

The goal of this project is to train a convolutional neural network (CNN) that can accurately classify skin lesions from dermoscopic images. We used ResNet-50 with weights pretrained on ImageNet as our backbone architecture (encoder) and replaced the last fully connected layer with a custom one for predicting the seven skin lesion classes (decoder/classifier). We trained several models, beginning with a simple baseline model where only the weights of the final fully connected layer were optimized, and progressively improved performance by addressing issues such as class imbalance and overfitting and by applying fine-tuning to leverage pre-trained feature representations for the skin lesion classification task.

Since we identified class imbalance as a key obstacle to achieving reliable performance, we explored multiple strategies for addressing class imbalance, including balanced batch sampling, class-weighted cross entropy loss, and focal loss. These approaches aim to ensure that minority classes contribute more effectively to the learning process. We found no clear performance advantage among the methods used to address class imbalance. Full fine-tuning of the network led to overfitting, and to mitigate this, we experimented with data augmentation, L2 weight decay, dropout regularization, label smoothing, and linear warmup. Model performance was evaluated using the balanced accuracy score.

We found that fine-tuning a pre-trained ResNet-50 network, combined with appropriate handling of class imbalance and strong regularization, improved classification performance to 74% balanced accuracy; however, overfitting remains an issue.

2 Related Work

Using deep learning to classify skin lesion images is not a new task. As early detection of skin cancer is crucial for effective treatment, the potential of such methods to assist clinicians in screening and diagnosis makes this a highly valuable and impactful area of research. A landmark study by Esteva et al. (2017) [2] demonstrated that convolutional neural networks (CNNs) can achieve dermatologist-level performance in the classification of skin lesions. In their work, the authors used the Inception v3 (GoogLeNet) CNN architecture pretrained on ImageNet and fine-tuned all network layers using transfer learning on a large skin lesion dataset consisting of 129,450 clinical images. For a three-class disease partition (benign lesions, malignant lesions, and non-neoplastic lesions), the CNN achieved an overall accuracy of $72.1 \pm 0.9\%$ (mean $\pm$ s.d.), exceeding the performance of two dermatologists. On biopsy-proven binary classification tasks, i.e. malignant versus benign lesions, the CNN achieved ROC-AUC values over 91%, matching or exceeding the performance of 21 board-certified dermatologists. In addition, t-SNE visualization of the 2,048-dimensional feature representations from the network's last hidden layer showed meaningful clustering of disease categories, indicating that the model learned clinically relevant and structured visual representations.

Since Esteva et al. (2017), many researchers have proposed improved architectures and training strategies to enhance accuracy and generalization for skin lesion classification. For example, in the study by Nawaz et al. [3], they compared their custom CNN with pretrained ResNet, EfficientNet, Inception, and MobileNet, and found that the specialized architecture combined with data augmentation and class weighting outperforms the generic pre-trained models, particularly for underrepresented lesion classes. Nevertheless, the task remains highly relevant and challenging today. Why is this task so difficult? The complexity and diversity of skin lesion images, that is, skin lesions vary greatly in appearance and even visually similar patterns can correspond to very different diagnoses, along with variations in image quality and lighting, and the scarcity and imbalance of labeled datasets, make it difficult for models to generalize reliably in real-world settings [2][4][5]. As a result, developing robust and clinically trustworthy systems for skin lesion classification continues to be an active and important area of research.

This continued research is made possible because larger datasets with high quality dermoscopic images and reliable diagnosis have been made available, for example the HAM10000 dataset [4] and the ISIC Archive [5]. But these datasets often show class imbalance, which needs to be properly addressed during model training.

Methods for addressing class imbalance can be divided into two main categories: data level methods and algorithm level methods [6]. Data level methods attempt to reduce the imbalance through various data sampling strategies such as undersampling (randomly removing samples from majority classes until all classes have the same number of samples) and oversampling (replicating randomly selected samples from minority classes). Data level methods thus change the class distribution in the training data. Algorithm level methods keep the training dataset unchanged and adjust the training algorithm for example assigning different weights to misclassification of training samples from different classes. In the study by Buda et al. [6] they found that oversampling is the method that works best with respect to multi-class ROC AUC to handle CNN training on imbalanced datasets.

In this report, we compare three different strategies to address class imbalance problem: (1) we used a minority class oversampling strategy where we implemented a balanced batch sampler to create batches with the same number of images from each class and augmenting all images in a batch; (2) we used a weighted cross entropy loss function; and (3) we used the focal loss function.

3 Methods

3.1 The Dataset and Evaluation Metrics

The HAM10000 dataset [4] consists of 10015 dermoscopic images of skin lesions which are classified into seven diagnostic categories. The dataset served as the training set for the ISIC 2018 challenge (Part 3) [5]. We downloaded the dataset from the ISIC Archive (link: <https://challenge.isic-archive.com/data/#2018>), along with the validation (193 images) and test (1512 images) datasets. The image labels have been confirmed by either histopathology (more than 50%), follow-up, expert consensus, or in-vivo confocal microscopy [4].

The dataset is extremely unbalanced; more than half of the images belong to one of the seven classes (Figure 1). Due to this class imbalance, it is not appropriate to use “standard” accuracy (overall proportion of correctly classified samples) for evaluating model performance, as it would be biased toward the majority class. Instead, we use balanced accuracy, which accounts for class imbalance by computing the accuracy for each class individually and then averaging these accuracies into a final score (this is equal to the average of recall or sensitivity obtained on each class) – ensuring that all classes contribute equally to the final score.

$$\text{Balanced Accuracy} = \frac{1}{C} \sum_{i=1}^C \frac{TP_i}{TP_i + FN_i}$$

This metric aligns with the “goal metric” used in the ISIC 2018 challenge [4]. In addition to balanced accuracy, we also consider and report precision and F1 (macro-average) scores (Appendix 2, Table 1) to provide a comprehensive assessment of model performance across all classes.

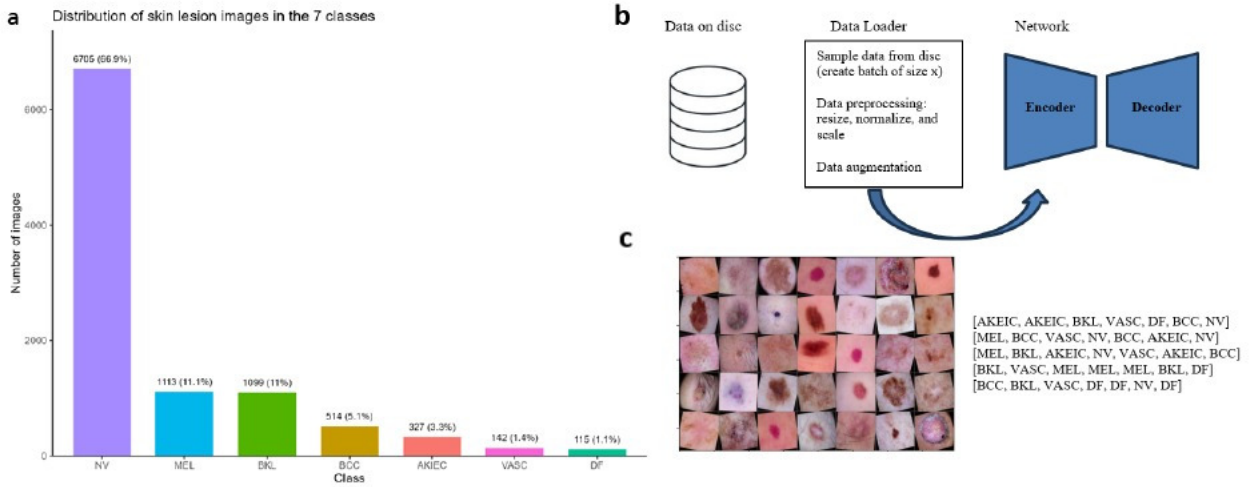


Figure 1. a) Histogram showing the number of images in each of the seven classes of the HAM10000 dataset (used as the training dataset in this project). b) Schematic illustration of batch construction by the data loader. c) Example of a balanced batch consisting of 5 images from each class with data augmentation applied. Dermoscopic images of skin lesions are classified into seven diagnostic categories: Melanoma (MEL); Melanocytic nevus (NV); Basal cell carcinoma (BCC); Actinic keratosis / Bowen’s disease (intraepithelial carcinoma) (AKIEC); Benign keratosis (solar lentigo / seborrheic keratosis / lichen planus-like keratosis) (BKL); Dermatofibroma (DF); Vascular lesion (VASC).

The number of images in each class in the training, validation, and test set can be found in Appendix 1.

3.1.1 Data Preprocessing

As the ResNet-50 architecture, pre-trained on the ImageNet dataset, requires input images of size 224×224 , all images were resized accordingly. The input pixel intensities were then scaled to the range $[0,1]$ and normalized using the mean and standard deviation values from ImageNet so that they are centered around zero. This ensures that the input images have the same properties as the images used to pre-train the ResNet-50 model, necessary for effective transfer learning. For all models other than the base model, data augmentation was also applied in real time (i.e., on-the-fly) during mini-batch creation for training (see section “Data Augmentation” for details).

3.2 Network Architecture

We used ResNet-50 [7] pretrained on ImageNet as our backbone architecture (encoder). ResNet is a deep convolutional neural network that uses residual blocks with skip connections, allowing the network to learn more complex features while mitigating the vanishing gradient problem commonly seen in very deep architectures. This makes it well suited and widely used for medical imaging tasks. The final fully connected layer was replaced with a custom one for predicting the seven skin lesion classes (decoder/classifier). Since we observed problems with overfitting, we did not expand the decoder with additional BatchNorm + ReLU + FC layers, as more weights increase the risk of overfitting.

3.3 Training strategy and Transfer learning

Training a convolutional neural network (CNN) from scratch, using randomly initialized weights, requires a very large dataset and takes a long time. To mitigate this, we rely on transfer learning, where we use a pretrained encoder which allows us to benefit from (generic) representations learned on large benchmark datasets, that can often be reused effectively. For the base model we used the pre-trained CNN encoder as a fixed feature extractor, and then simply trained a classifier consisting of a single fully connected layer on the extracted features using the data set. However, because the characteristics of dermoscopic skin lesion images differ considerably from those in ImageNet - the dataset on which our encoder was originally trained - it is reasonable to question how well these generic features generalize to our domain [3]. For this reason, we chose to perform full fine-tuning, allowing all model parameters to be optimized on the skin lesion dataset. This comes with the risk of overfitting, and we thus applied several regularization techniques in order to reduce overfitting.

We trained the networks using stochastic gradient descent with momentum set to 0.9. When doing experiments, we decided to train all models for 10 epochs. Is the model done training after 10 epochs? One could argue that the validation loss still goes down slightly and that the model should train for more epochs, but because training takes quite some time and we wanted to do more experiments, we trained for only 10 epochs. Also, the balanced accuracy metric has largely stabilized and does not seem to improve much by training longer (see the learning curve figures in the “Results” section). After experimenting with strategies to handle class imbalance and overfitting, we chose the best performing model (on the validation set) and trained for 20 epochs and experimented with a learning rate schedule.

For all models we applied early stopping, that is, we stopped training when the validation loss no longer decreased, and we kept the model with highest balanced accuracy. At the point where the validation loss begins to plateau or increase while the training loss continues to decrease, the model starts to overfit the training data. Since we want a model that generalizes well to new unseen data, not just fit the training data, early stopping is essential.

3.3 Experiments

3.3.1 Base model

For the base model, we trained only the final fully connected classifier layer while keeping the remaining network parameters frozen. We used the cross-entropy loss function to quantify the discrepancy between the predicted class probabilities and the ground-truth labels. After training for five epochs and evaluating on the validation set, we observed (as expected) that class imbalance is a huge problem (see Figure 2 with confusion matrix in “Results” section). Although the training was not fully converged, this preliminary experiment was sufficient to reveal the imbalance-related issues in the baseline setup.

3.3.2 Dataloader and Batch size

We implemented our own dataloader to create batches of 35 images (see Figure 1b). The batch was either created through random sampling (base model and models using a weighted loss function) or by using a balanced batch sampler that selects five images from each of the seven classes. Data augmentation was applied on a per-batch basis before feeding the images into the network for training.

3.3.3 Data augmentation

We applied data augmentation during training for all models except the base model. Data augmentation artificially increases the number of training samples, which is particularly important in our setting because we oversampled minority classes to address class imbalance. Additionally, augmentation acts as a regularization technique that helps reduce overfitting.

All augmentations were applied on the fly when creating a batch during training. Since lesion morphology and color patterns are essential diagnostic features, it was important that augmentation strategies preserved these characteristics. We used random cropping and scaling while ensuring that 90–100% of the original image content was retained to avoid excluding the lesion. We also applied random horizontal and vertical flips, small

random rotations, and mild color jitter. The intensity of the color jitter was kept deliberately low to simulate natural variations in lighting without altering color cues that are important for clinical interpretation (Figure 1c).

3.3.4 Strategies to address class imbalance

We experimented with three strategies to address and mitigate the effects of class imbalance: (1) a balanced batch sampler, (2) a weighted cross-entropy loss function, and (3) focal loss. Our goal was to make full use of the available dataset and not downsample the majority classes to match the size of the smallest class.

3.3.4.1 Balanced batch sampler

The balanced batch sampler addresses class imbalance by constructing batches that contain the same number of images from each class. With a batch size of 35, each balanced batch contains five images from each of the seven classes. This approach ensures that the network frequently sees minority class images and that each class contributes equally to every training step. The potential disadvantage is that balanced batches do not reflect the true class distribution and thus the model sees an artificially uniform distribution during training, which may lead to poorer performance when evaluated on data with natural imbalance. Also the reuse of minority class samples (minority classes have fewer examples, so they appear more often in batches), increases the risk of overfitting unless strong data augmentation is applied.

3.3.4.2 Weighted cross entropy loss function

Weighted cross entropy handles class imbalance by assigning higher loss weights to classes that contain fewer images (underrepresented classes) and lower weights to classes with more images (overrepresented classes). We compute class weights as the reciprocal of the class frequency and normalize them by the number of classes so the average weight is close to one. These weights are then incorporated into the cross entropy loss function, so that the loss is scaled according to the class distribution, ensuring that the model is penalized more for errors on rare classes, thereby reducing bias toward majority classes. Formula for computing class weights: $w_i = N / C \cdot n_i$ (N is the total samples, C is the number of classes, n_i is the number of samples in class i).

3.3.4.3 Focal loss

Focal loss [8], originally introduced as a solution for class imbalance in one-stage object detectors, uses two parameters which together modulate the loss function to account for both class frequency and difficulty of the examples. It builds upon the basis of weighted cross entropy to target minority classes. For multiclass classification, a weighing factor α (alpha) assigns a weight to each class based on its inverse frequency in the dataset, improving the contribution of the minority classes [9]. The weights are normalized by their mean and clamping ($\alpha_{\max}=2.0$) to prevent excessively large weights from dominating the loss. The weighting factor is accompanied by a focussing parameter γ (gamma), which down-weights the influence of well-classified (easier) examples on the loss function, improving the predictions of misclassified (harder) examples over time. The default setting of $\gamma = 2$ was used as proposed in the original work by Lin et al. Formula for computing focal loss: $FL(p_i) = -\alpha_i \cdot (1 - p_i)^\gamma \cdot \log(p_i)$, where p_i - predicted probability of the true class, α_i - class weighing factor for class i, γ - focussing parameter.

3.3.5 Regularization – strategies to address overfitting

We observed overfitting early in the training process (see “Results” section), indicating that the model began to memorize the training data rather than learning representations that generalize well to unseen samples. To mitigate this issue, we applied several regularization strategies. We have previously discussed data augmentation and early stopping. The additional methods are described below.

We applied weight decay by adding an L2 regularization term to the cross-entropy loss, with a regularization coefficient of $\lambda = 1 \times 10^{-4}$. This parameter controls the trade-off between minimizing the data loss and minimizing the model parameters. By penalizing large weights, weight decay encourages simpler models (smaller weights) and reduces the risk of overfitting by preventing the network from fitting noise in the data.

Dropout was applied to the final fully connected layer during training. Neurons were randomly set to zero during each forward pass with a dropout probability of $p = 0.5$. This forces the network to learn redundant and more robust feature representations, thereby improving generalization.

While these standard regularization techniques reduced overfitting, they were not sufficient to eliminate it entirely (see “Results”). To further improve validation performance, we tried to implement label smoothing. The ISIC 2018 challenge provides metadata indicating how each label was confirmed. Since histopathology represents the gold standard, label smoothing was not applied to samples with histopathological confirmation. For the remaining samples, label smoothing was used to replace hard one-hot labels with soft probabilistic targets. We did smoothing = 0.1, which means that the true class probability becomes $1 - \text{smoothing} (0.9)$ and the remaining 10% of the probability mass is distributed equally over the other classes. This reduces overconfident predictions and improves robustness to potential label noise. However, we did not observe any performance gain from applying label smoothing, and it is therefore not further discussed in this report.

3.3.6 Learning rate schedule

When performing experiments, the initial learning rate was $1 \cdot 10^{-3}$ and was reduced by a factor of 0.1 every seven epochs. The idea is that this schedule allows the model to first explore the loss landscape before converging toward a local minimum. In practice, we saw that the validation loss plateaued early in the training process, and we suspected that this might be due to early overfitting and too high learning rate. Thus we implemented linear warmup and decreased the max learning rate to $5 \cdot 10^{-4}$. Linear warm-up is a training technique where the learning rate starts very small and is gradually increased to its target value over the first few epochs. This helps stabilize early training. Linear warmup was followed by cosine learning rate decay.

4 Results

The base model clearly demonstrated a class imbalance problem, as evidenced by the confusion matrix generated from validation set predictions (Figure 2): the model predominantly predicted the majority class (NV) and failed to assign any samples to the two classes with the fewest training images (DF and VASC). Class imbalance was thus the first problem we sought to solve. We experimented with three strategies to address class imbalance: balanced batch sampler, weighted cross entropy loss, and focal loss (described in Methods). Combining full network fine-tuning with these strategies increased the balanced accuracy on the validation set from 0.38 for the base model to approximately 0.84. The resulting confusion matrices presented in Figure 2 demonstrate that all three approaches substantially reduced the negative impact of class imbalance compared with the base model and improved performance on the minority classes. The balanced accuracy, specificity, and F1 score were similar across all three approaches (Appendix 2, Table 1).

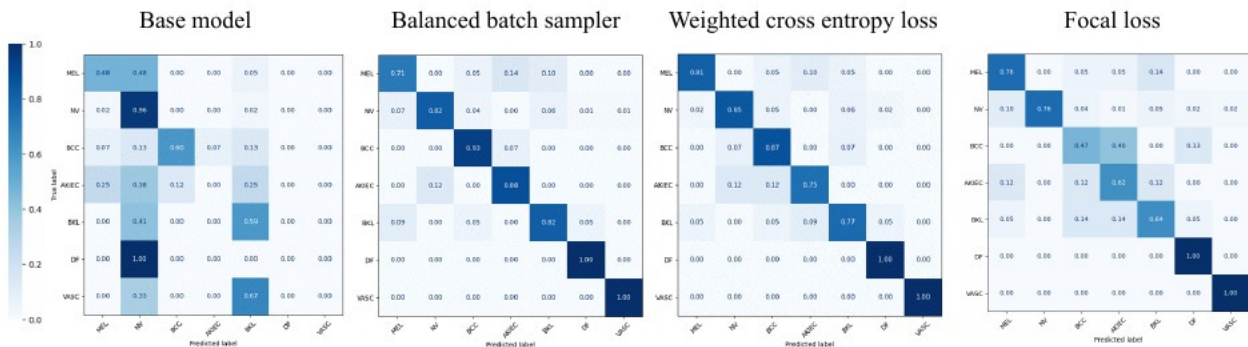


Figure 2. Confusion matrices for validation set predictions comparing the base model with the three strategies for addressing class imbalance.

As none of the methods for addressing class imbalance seemed superior, we selected the balanced batch sampler for further experiments and model improvements.

Analysis of the learning curves - showing loss, balanced accuracy, specificity, and F1 score for both training and validation data over epochs (we only show loss and balanced accuracy curves in this report) - revealed that full fine-tuning of the network leads to overfitting early in the training process (Figure 3a). While the training

balanced accuracy continued to increase and the training loss kept decreasing, the validation balanced accuracy declined and the validation loss plateaued, indicating that the model starts to memorize the training data, learning training-specific patterns that do not generalize to unseen data. Overfitting was therefore identified as the second major challenge.

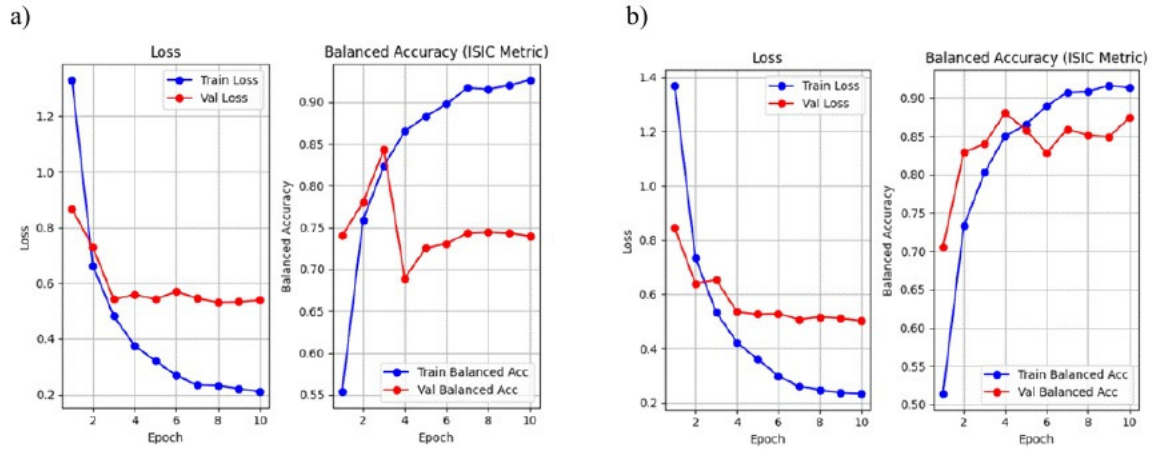


Figure 3. a) Learning curves for the model trained using the balanced batch sampler (with data augmentation) and no additional regularization. b) Learning curves for the model trained using the balanced batch sampler (with data augmentation) and with dropout ($p = 0.5$) and L2 weight decay ($\lambda = 1 \times 10^{-4}$).

Various regularization techniques (described in Methods) were applied to reduce overfitting: data augmentation (already part of the balanced batch sampler), dropout, and L2 weight decay. These methods narrowed the gap between training and validation performance, but they did not eliminate overfitting entirely (Figure 3b). Applying these regularization techniques improved the balanced accuracy on the validation set by a few percentage points, reaching performance ranging from 0.86 to 0.88.

The performance of the models evaluated to address class imbalance and overfitting is summarized in Table 1.

The validation loss and balanced accuracy still reached a plateau very early in the training process (Fig. 3b), which suggests that the learning rate of $1 \cdot 10^{-3}$ may be too high. We also suspected “early overfitting” where early examples have too much influence which results in unstable or biased learning in the very first training iterations before the model has seen enough diverse data. We therefore tried different learning rate schedules and extended training to 20 epochs for the final model. The combination of linear warm-up (1000 steps, approximately 3.5 epochs) to a peak learning rate of $5 \cdot 10^{-4}$ followed by cosine learning rate decay made the training process smoother but did not increase the overall validation performance (balanced accuracy = 0.87) (Figure 4). Precision (0.79) and F1 score (0.82), however, showed a slight increase.

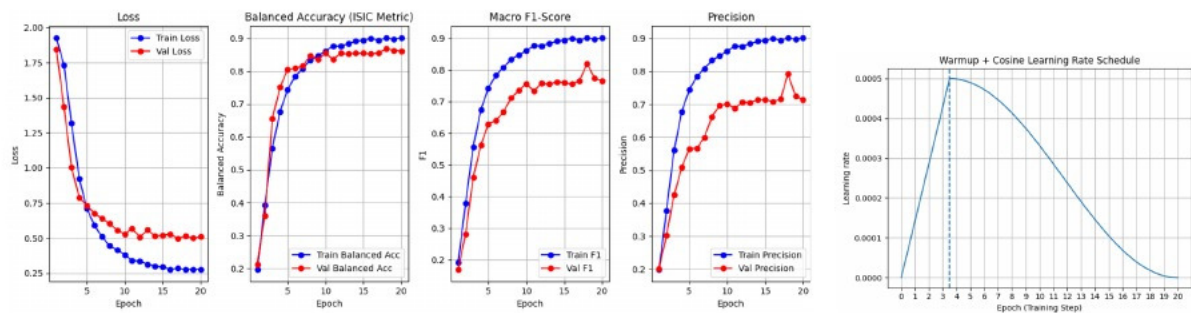


Figure 4. Left: Learning curves for training with linear warm-up followed by cosine learning rate decay. Right: Corresponding learning rate schedule.

Although the validation loss continued to decrease slightly toward the end of training, indicating that the model might benefit from longer training (additional epochs), training was not extended further due to time constraints and Google Colab GPU usage limits.

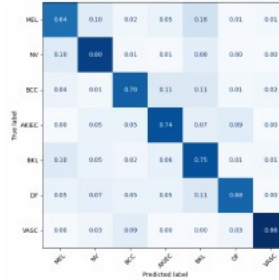
Finally, the best-performing model on the validation set (balanced batch sampler with data augmentation, dropout, L2 weight decay, linear warm-up, cosine learning rate decay, and 20 training epochs) was evaluated on

the test set. On the test set we obtained a balanced accuracy of 0.74, precision of 0.66, and F1 score of 0.69. Performance on the test set was lower than on the validation set, indicating that overfitting remains an issue and that the model does not fully generalize to unseen data (Figure 5).

a) Performance metrics - test set

	precision	recall	f1-score	support
MEL	0.48	0.64	0.55	171
NV	0.96	0.80	0.87	909
BCC	0.73	0.70	0.71	93
AKIEC	0.44	0.74	0.56	43
BKL	0.58	0.75	0.65	217
DF	0.70	0.68	0.69	44
VASC	0.75	0.86	0.80	35
accuracy			0.76	1512
macro avg	0.66	0.74	0.69	1512

b) Confusion matrix - test set



c) Top / bottom images

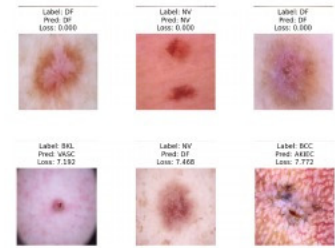


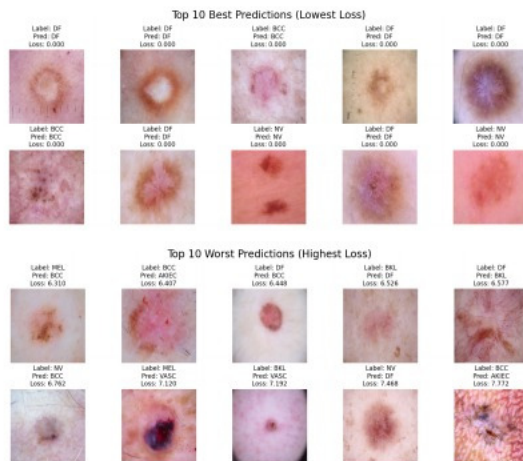
Figure 5. Evaluating performance of the final model on the test set. a) Performance metrics. b) Confusion matrix. c) Top / bottom images with lowest / highest loss.

5 Discussion

Our model does not perform as well as the top models in the ISIC 2018 challenge, where the winners achieved a balanced accuracy of 0.885 [5]. Nevertheless, the obtained performance is comparable to that reported in the literature for similar CNN-based skin lesion classification tasks [2][5]. Of particular concern is the model's poor performance on the clinically most critical class, melanoma, which is malignant. Compared with the other classes, the model achieves noticeably lower performance for melanoma, with a precision of only 0.48, indicating a high rate of false-positive predictions. This reduced reliability in detecting the most dangerous lesion type represents a significant limitation of the model from a clinical perspective.

As the model's performance decreases when evaluated on the test set, overfitting remains an issue. We therefore investigated whether the model relies on specific features from the training data when making predictions. To gain further insight into what the model has actually learned and when the model succeeds and fails, we analyzed the top and bottom 10 images in the test set with the highest or lowest loss values, respectively (Figure 6a). In addition, Grad-CAM heatmaps were used to visualize which regions of the images the model attends to (focuses on) when predicting the skin lesion class (Figure 6b). We did not observe any clear visual patterns distinguishing correctly classified images from misclassified ones; however, some of the lesions which the model fails to classify, appear to lack a well-defined lesion boundary. The Grad-CAM heatmaps indicate that the model primarily focuses on the lesion area during prediction rather than on surrounding skin or artifacts such as hair, which suggests that the model attends to relevant image features (Figure 6b).

a)



b)

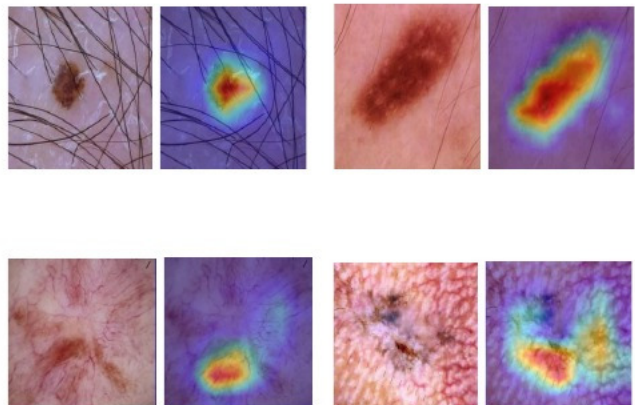


Figure 6. a) Top / bottom images with highest / lowest loss. b) Grad-CAM heatmaps of selected images.

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Appendix 1 - Number of images in each class in the training, validation, and test set

Label	Training data	Validation data	Test data
NV	6705	123	909
MEL	1113	21	171
BKL	1099	22	217
BCC	514	15	93
AKIEC	327	8	43
VASC	142	3	35
DF	115	1	44

Appendix 2 - Table 1 showing an overview of models and comparing performance on validation set

Model	Strategy to solve class imbalance	Data Augmentation	Regularization (L2 weight decay and dropout)	Balanced Accuracy (average recall)	Precision (macro average)	F1 (macro average)
Base	No	No	No	0.38	0.42	0.39
v1.1	Balanced batch sampler	Yes	No	0.84	0.67	0.72
v1.2		Yes	Yes	0.88	0.66	0.74
v1.3		Yes	Yes + label smoothing	0.85	0.66	0.72
v2	Weighted cross entropy loss function	Yes	Yes	0.86	0.69	0.73
v3.1	Focal loss with $\alpha = 1$ (2step)	Yes	Yes	0.84	0.78	0.79
v3.2	Focal loss with $\alpha = \text{class weights}$ (ft directly)	Yes	Yes	0.75	0.51	0.57
v4	Focal loss $\alpha = 1 + \text{balanced batch sampler}$ (ft directly)	Yes	Yes	0.87	0.79	0.82

Appendix 3 - links to source code (Google Colab notebooks)

source code 1 main skin lesion classification:

<https://colab.research.google.com/drive/>

[?usp=sharing](#)

source code 2 label smoothing:

<https://colab.research.google.com/drive/>

?usp=sharing

source code 3 focal loss:

<https://colab.research.google.com/drive/>

[?usp=sharing](#)